Tools for meta-analysis

G0B75A Meta-analysis
Master of statistics

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Tools for meta-analyses

a. checklists and guidelines
b. software for reference management
c. software for systematic reviews
c. (other) GenAI / LLM-t

https://casp-uk.net/casp-tools-checklists/

Critical Appraisal Checklists

We offer a number of **free downloadable checklists** to help you more easily and accurately perform critical appraisal across a number of different study types.

The CASP checklists are easy to understand but in case you need any further guidance on how they are structured, take a look at our guide on how to use our CASP checklists.

- Systematic Reviews with Meta-Analysis of Observational Studies
- Systematic Reviews with Meta-Analysis of Randomised Controlled Trials (RCTs)
- Randomised Controlled Trial (RCT) Checklist
- Systematic Review Checklist
- Qualitative Studies Checklist
- Cohort Study Checklist
- Diagnostic Study Checklist
- Case Control Study Checklist

- CASP Checklists
- How to use our CASP Checklists
- Referencing and Creative Commons

Section A: Is the basic study design valid for a systematic review?	
1. Did the systematic review address a clearly formulated research question?	☐ Yes ☐ No ☐ Can’t Tell

CONSIDER:
Did the researchers state a research question and a null hypothesis? For a systematic review of RCTs, a research question can be ‘formulated’ in terms of the PICOT(S) framework:

- *Population*
- *Intervention*
- *Comparator*
- *Outcome/s and Outcome measures*
- *Time, e.g., study timeframe, or follow-up intervals*
- *Setting*

2. Did the researchers search for appropriate study design(s) to answer the research question?	☐ Yes ☐ No ☐ Can’t Tell

CONSIDER:
If the research question is concerned with the efficacy of an intervention, the RC
study design for a systematic review. The most common type of RCT is the paral
individuals are randomised to study groups; other methods of randomisation, h

The ROBIS tool for the assessment of the quality of systematic reviews

https://www.bristol.ac.uk/population-health-sciences/projects/robis/robis-tool/

https://www.prisma-statement.org/prisma-2020-checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation	
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study interve comparing against the planned groups for each synthesis (item #5)).	
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summa conversions.	
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was perfo model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis	
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases)	

PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	
Study characteristics	17	Cite each included study and present its characteristics.	
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	
	23b	Discuss any limitations of the evidence included in the review.	
	23c	Discuss any limitations of the review processes used.	
	23d	Discuss implications of the results for practice, policy, and future research.	
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the revie	
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the rev	
Competing interests	26	Declare any competing interests of review authors.	
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data studies; data used for all analyses; analytic code; any other materials used in the review.	

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Endnote, **Mendeley**, **Zotero**:

- Import the ‘hits’ from the database searches
- Automatically remove duplicates
- Add notes
- Attach files (e.g., full text in pdf) to references
- Make reference list for your report

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- **EPPI-reviewer** (https://eppi.ioe.ac.uk/): free, web-based tool. Reference manager, but also for data management (creating codes, coding with multiple coders, divide papers over coders, calculation of inter-coder agreement, …)
- **Rayyan** (https://www.rayyan.ai/) & **Covidence** (https://www.covidence.org/): Idem but AI-powered, e.g., by highlighting text (probably) related to the elements of PICO; learn relevance of studies based on included studies; …
- **Elicit**: https://elicit.com/ (automatic screening; automatic data extraction; …)

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- Perplexity AI: https://www.perplexity.ai/ (search for answers, with references to sources)
- Consensus AI: https://consensus.app/search/ (search in peer-reviewed academic literature, gives references, evaluates how much consensus there is)
- Semantic scholar: https://www.semanticscholar.org/ (discover and summarize academic papers)

A new field!

First promising results of using LLMs for study searching, study selection, data extraction and research synthesis.

Luo, R., Sastimoglu, Z., Faisal, A. I., & Deen, M. J. (2024). Evaluating the efficacy of large language models for systematic review and meta-analysis screening. *medRxiv*, https://doi.org/10.1101/2024.06.03.24308405

Oami, T., Okada, Y., & Nakada, T. (2024). Performance of a large language model in screening citations. *JAMA Network Open*, *7*, e2420496. https://doi.org/10.1001/jamanetworkopen.2024.20496

Susnjak, T., Hwang, P., Reyes, N. H., Barczak, A. L., McIntosh, T.

S. (2024). Automating research synthesis with domain-spe

language model fine-tuning. *arXiv*. https://doi.org/10.4855

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Good luck and take care!